

Glen A. Dsouza

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Engineering Portfolio: flame1113.github.io/Portfolio - full write-ups, 3D models, schematics & build logs for every project below
U.S. Citizen

EDUCATION

California Polytechnic State University, San Luis Obispo

Expected May 2030

B.S. Electrical Engineering - College of Engineering

TECHNICAL PROJECTS

Power Wheels - Centralized Vehicle Control Unit (CVCU)

June 2026 - August 2026

Designer & Integrator

- Designed a custom PCB as the centralized control hub for a modified Power Wheels ride-on, integrating an ESP32-WROOM-32U for telemetry, RFID access control, an OLED UI, and TPA3118-based audio amplification.
- Implemented AO3400 N-channel MOSFET propulsion switching with SPDT relay isolation and 30A motor terminals; populated the board via reflow and precision hot-plate soldering.
- Programmed WS2815 addressable LED lighting effects and built a self-hosted, mobile-responsive web UI on the ESP32 for real-time dashboard control of lighting, audio, and RFID status over Wi-Fi.
- Diagnosed and resolved recurring SPI/serial faults traced to flux contamination and unshielded signal runs using an oscilloscope and multimeter, improving RFID and communication reliability.

Custom Mini-Z 1:28 RC Platform

Sep 2025 - Dec 2025

Designer & Fabricator

- 3D-printed the chassis for a 1:28-scale RC platform paired with a custom KiCad controller PCB (ATtiny85, DRV8833PWP) designed under sub-millimeter tolerance constraints.
- Integrated propulsion and steering servo subsystems by reverse-engineering the radio receiver's PWM signal output and deriving timing-to-motion conversion math to calibrate throttle and steering response; independently assembled, soldered, and validated electromechanical performance.

Facial Recognition Operated Door Lock

June 2023 - July 2024

Project Developer

- Built an ESP32-based access control system with a helix keypad, 5V servo actuation, LCD feedback, and buzzer status signaling to enable remote and password-based entry.
- Developed firmware in Arduino C++ for password validation and lock-state management, and integrated an ESP32-CAM with a Python/OpenCV facial recognition pipeline for camera-based access.

ENGINEERING ORGANIZATIONS

Prototype Vehicles Laboratory (PROVE), Cal Poly SLO

Aug 2026 - Present

Member

- Designed a compact two-layer Altium PCB for a full-scale vehicle steering-controller interface supporting PROVE's autonomous, fully self-driving vehicle development; integrated fused 12 V input, regulated 5 V/3.3 V rails, an ESP32-WROOM-32D, steering-angle sensing, and TCAN337 CAN communication.
- Added a USB-C/CH340C interface, reset/boot controls, and status indicators to support firmware flashing, serial debugging, and board bring-up; completed design-rule checks and documentation.

IEEE Cal Poly Student Branch

Aug 2026 - Present

Member

- Participated in hands-on workshops covering PCB design and layout, soldering, and circuit assembly practices.
- Developed practical familiarity with oscilloscopes and bench-test workflows for signal inspection, measurement, and circuit troubleshooting.

EXPERIENCE

Kumon - Thousand Oaks, CA

May 2022 - Dec 2025

Paid Tutor

- Evaluated and graded math and reading assessments for students across grade levels K-12.
- Delivered targeted instructional support to students requiring additional concept reinforcement.
- Opened and independently managed the tutoring center in the owner's absence, overseeing daily operations and maintaining a clean, organized learning environment.

SparkPrintz - Remote

June 2024 - Aug 2024

Sales & Marketing Intern

- Cold-emailed school administrators and faculty to market apparel-printing services for an early-stage startup.
- Conducted market research and authored business analysis write-ups on company positioning.
- Participated in professional development panels and discussions with Ivy League graduates, gaining insight into career planning and industry pathways.

TECHNICAL SKILLS

Programming: C++, Python, Java, JavaScript, HTML/CSS, SQL, Git, GitHub

EE / Design: Altium, KiCad, AutoCAD, PCB Layout, Schematic Capture, SMD Assembly, Circuit Design & Debugging, Power Electronics

Embedded: ESP32, ATtiny85, Arduino, PWM Motor Control, I2C, SPI, UART, Signal Integrity

Test & Fab: Oscilloscope, Multimeter, Bench Power Supply, Fusion 360, FDM 3D Printing, Reflow Soldering, Enclosure Design